

Dopamine, via its receptors, plays a vital role in the maintenance of blood pressure by modulating renal sodium transport. However, the role of the D₄ dopamine receptor (D₄ receptor) in renal proximal tubules (PRTs) is still unclear. We hypothesized that activation of D₄ receptor directly inhibits the activity of the Na⁺-K⁺-ATPase (NKA) in RPT cells.